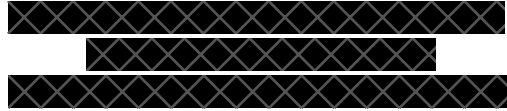


AquaPath: Closest swimming places in the City of Copenhagen



2025-05-23

Abstract

As part of the Sustainable Development Goals, the UN includes requirements for making cities more human-livable and sustainable. In the efforts to comply, the city of Copenhagen is developing plans for, among other things, making the city greener, more accessible, with better biking and swimming infrastructures. This effort requires having knowledge of the current situation, by having systems in place to analyse the water quality, collect the results, and report them to the citizens, so they can make better decisions when using the provided facilities. On top of that, it is necessary to assess the current accessibility of the swimming places, so that informed decisions can be made when choosing where to add new ones. Our product is a prototype of a shortest path map system for users to decide where to go swimming, based on information such as water quality and distance, and an accessibility map of the city in terms of walking and biking distance to the closest bathing place. The result is a working product with the necessary features, based on an interactive map for the user. We discuss the limitations of the product and the possible future work, but argue for the usability of the current product for decision-making.

1 Introduction

The Sustainable Development Goals (SDG) defined by the Department of Economic and Social Affairs of the United Nations [1, p.15] include requirements for cities to become more livable and accessible to the inhabitants. The following report will focus specifically on Sustainable Cities and Communities, number 11, and Clean Water and Sanitation, number 6. The municipality of Copenhagen has been working for more than a decade towards making the city more and more livable, offering better amenities and a more human-centered environment, where people can meet, enjoy the space, and be part of the city's everyday [2]. Part of the plan is improving accessibility to green areas and offering a better swimming experience in the harbor [3]. The latter is the focus of this project.

Part of the allure of the city of Copenhagen is the possibility of swimming in the canals and harbours of the city, all year round. Winter and summer bathing are a part of the popular activities to do in the city, and very often, one might find people swimming, no matter the weather. Ensuring the quality of the water is a challenging task, as it varies from day to day due to heavy rainfalls, sewage draining, and marine traffic [4]. For this reason, there needs to be systems in place to monitor and report water quality in real time, so that people are informed of the water conditions when deciding if, and where, they want to swim. There exist multiple tools that allow the user to check the water quality. The Copenhagen Municipality has invested

efforts and resources into a series of platforms that allow users to check current and predicted water quality [4], get information about bathing in Copenhagen [5], and get a comprehensive view of the bathing water quality all around Denmark [6]. Within our current knowledge, there is a system that allows users to find the bathing station closest to them, while also informing on the water quality measurement, which is the app version of the overview of bathing water quality in Denmark [6]. However, the app does not provide a path from the location to the swimming place. Thus, our contribution will focus on integrating a visual shortest walk path from any location the user chooses to the closest swimming facilities, coupled with water quality information.

More formally, we have developed an interactive map of the city of Copenhagen that allows the user to select where they are directly on the map. The map then shows the path to the closest bathing station, with relevant information. On top of that, we have created an accessibility map to bathing places from every part of the city. This map is used as support for the analysis of the quality of life based on how accessible the stations are to users. This map could be used by the municipality in the effort to improve the city, and by individuals for choosing, possibly, areas that are more suited for their needs.

2 Background

As mentioned in the introduction, one of the Sustainable Development Goals is centered around making cities and human settlements inclusive, safe, resilient and sustainable [1, p.24] point 11.3 specifically requires to *"enhance inclusive and sustainable urbanization and capacity for participatory, integrated and sustainable human settlement planning and management in all countries"*. This statement can be interpreted as creating human settlements that allow for inhabitants to participate in the life of the place, making it sustainable both in terms of the environment and human usage. This is the focus of the city of Copenhagen when improving the space for people to use. They aim at improving accessibility to green areas, ensuring the best conditions for biking and swimming in the harbour [3], among other things. Practically, however, there are many ways to consider improvement in the urban area. Ensuring clean spaces, with working facilities and good resources, is a great part of the work. On the other hand, the facilities need to be accessible to all within a reasonable time, otherwise the disparity of accessibility risks increasing socio-economic disparity, and degrading the overall quality of life. Thus, creating equal opportunities for access to services such as bathing stations in the harbour is an integral part of increasing parity among the city's inhabitants [7].

As an initial approach to solve this problem, we present and argue for the need of a system that provides the shortest path from wherever the user is, to the closest bathing place in the city. As previously mentioned, there exist platforms that show the closest swimming stations, with relative water quality data. However, to get the best path to reach the place, the user has to use another platform. By integrating the two systems, we can offer a simple solution that allows the user to make an informed decision on where to go swimming, based on preferred bathing place, current position, or even other positions. The user is not bound by current location, like in other tools we could find, but can choose wherever on the map, which can be of use when deciding, for example, if they would consider moving to a different part of the city, or for tourists deciding where to stay. Having access to the necessary natural amenities can be an integral part of residents' happiness [8], and in particular, having access to waterfronts can be majorly beneficial for mental health [9]. Not everybody has the same needs, so it is important to make a distinction between accessing waterfronts and accessing swimming places, but the conceptual background is similar. Thus, providing an interactive tool for assessing distance to

specific swimming places, both for the current location and future planning, can be beneficial for the continuous development of the city.

However, accessibility isn't just defined by the shortest path to a location. Especially after the COVID-19 global pandemic, the concept of accessible amenities and essential services through the devise of 15-minute cities has become widespread. Initially devised in 2016, and more consistently developed after 2021 [10], the 15-minute city represents the accessibility of the city in terms of essential services and amenities, so that people can reach them in 15 minutes or less via transport that is not polluting (walking and biking). With the use of visual tools such as isochrone maps (representing the area accessible from a point within a certain time threshold) [11], we can represent a city's accessibility as a visual overview, easily comparable.

In 2024, Bruno, Monteiro Melo, Campanelli, *et al.* published a paper on the topic with an online platform that allows the user to see many cities worldwide through the 15-min map lens [12]. Copenhagen is mentioned as a city that is showing a well-optimised, homogeneous distribution of services. They state that the need for re-allocation of points of interest is quite low (around 5.2%), and the effectiveness of the current urban design is quite high. On top of that, they analyse the population density and devise solutions to offer amenities based on the number of people using them, to make the city become a 15-minute city, for the cities that necessitate it. They discuss the introduction of the notion of a city optimized for amounts of opportunities, on top of transit time, so that the residents have equal opportunities in the city.

We believe our project can contribute to this particular point. While they devise a map based on points of interest, from the ones provided by OpenStreetMap, we only focus on bathing stations in the city. Clearly, the bathing stations need to be in a space where there is water, so we can expect the analysis to be biased against places that have a high distance from the sea. Nonetheless, due to urban planning, there might be areas that are close to the water but far from bathing stations. This point can highlight the development of new and more accessible stations for people in the city.

3 Data

3.1 Data Acquisition

As presented in the introduction, there are already some platforms that present bathing station water quality. However, none of these sites share their data, that we could find, so we had to download data from the source, which is the Danish environmental portal [13]. The portal is an open source platform for collection and sharing of data, owned by the danish state, and containing data collected by over 150 organizations. The goal is to allow users to find environmental data, and use it for analysis, protection and conservation of the national environment. Considering the purpose and the owners we are comfortable trusting it as a data source.

3.2 Data Processing

The dataset is composed of information merged from two files. The first dataset contains water quality measurements for the different bathing sites around Copenhagen. The data is downloaded from [14], which is part of Danmarks Miljø portal and allows the user to select an area on the map for which the data should be displayed and downloaded. This data can also be found here: [15]. By making a geometry around the area of interest (Copenhagen city), we were able to highlight the places where there have been water quality measurements. These places included both bathing sites and non-bathing sites, so the first step in the data cleaning was to restrict the data to only bathing places, which returned 35 individual sites within 15km of Copenhagen

city. The data contains information on the water quality measurements from 1990 up until April 2025, when we downloaded the data. We decided to only analyse the water quality for the latest measurement for each of the 35 bathing places. Some of the measurements were taken recently (April 2025), and some were much older. This information is surprising, considering that the bathing stations are still open. Visually assessing the position of the places on the map, we realised that the older measurements were from places that have other bathing stations nearby, which suggests that the measurements for these places are done only at one of the stations. Finally, we created a boolean flag for whether the measurement passes the standard of water quality, which is also defined within the dataset. This process allows us to display reliable information to the user, based on pre-set standards. The parameter measured is also presented.

The resulting dataset is composed of the name of the bathing station, the date of the latest measurement, the Kommune of the bathing station, the GeoZone location, X and Y coordinates and the quality test information. The coordinate system of the data appears to be in EPSG:25832 [16], according to the data source [15]. As the specific coordinate system is not very clear, we did some analysis on the most common ones used in Denmark, and tested the results by plotting the data. The plot shows that the hypothesis was correct, and the coordinate system gave the right results on the map.

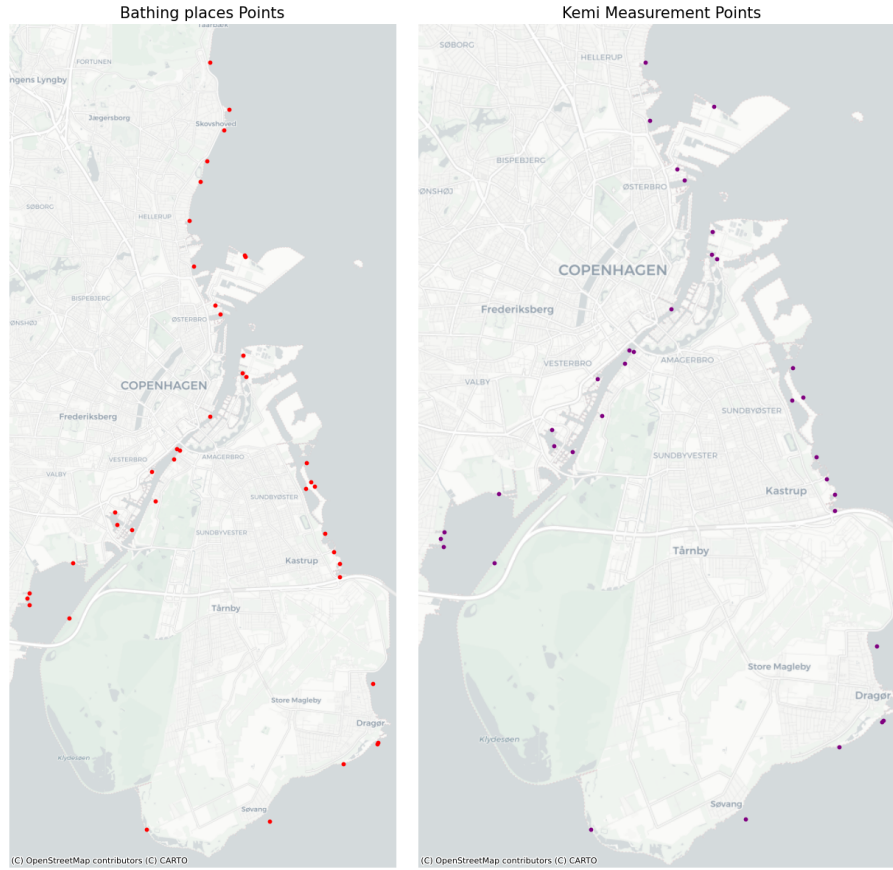


Figure 1: Plot of points from the two dataframes to compare accuracy of coordinates

The second data source we used is related to the bathing station sites. It was downloaded

from Badevand stamdata [17], and it contains information about the bathing stations in different municipalities around Denmark, such as name, latitude and longitude, water types, and the municipality responsible for it. As the data is related to the whole country, we filtered for the same municipalities that were defined in the previous dataset, as the ones with bathing places max 15km from the city. Any irrelevant column for our analysis was dropped to allow better readability. For each place we had information on, the name, latitude, longitude and water area (e.g., marine). For this data, the CRS was specified to be EPSG:25832 [17]. However, when plotting the data, we noticed that the system was incorrect. Thus we tried with the most common one for the type of lat/lon data, which is EPSG:4326 [18], which proved to bring the correct results.

As mentioned, both datasets have geo-information, X/Y coordinates for the chemical data, and latitude and longitude for the bathing stations data. We thus decided to merge the two datasets on their coordinates. Again, we assessed the correctness of the results by plotting them on the map. We used the exported data for the rest of the project. The manipulation of the data is done in the notebook "data_cleaning.ipynb" [19].

4 Method

Our project consists of the frontend and backend. The frontend allows users to interact with a map to:

- View the shortest path to the nearest swimming place
- See locations and water quality information for swimming places
- Visualize the accessibility of swimming places using hexagonal isochrone maps

The backend is responsible for computing and serving all necessary data to the frontend. In addition, we developed several stand-alone data extraction and preprocessing scripts to support the main application.

In this project, the main idea is to calculate the shortest path to swimming places in both steps (calculating and drawing the shortest path and a hexagonal isochrone map). Therefore, we needed a network of the map with nodes and edges where we can calculate the distance between nodes through edges. Although Routing APIs like Google were an option, we chose not to rely on external services. Another option was to use Euclidean distance, but this approach is unrealistic as it overlooks road layouts, barriers, and detours.

In order to retrieve the network data from map, we first downloaded the network graph within a 15 km radius around Copenhagen as *graphml* file using *OSMnx* [20]. Since downloading the network on every run is computationally expensive and the network doesn't change frequently, we opted to download it once and load the saved map instead.

Figure 2 illustrates walking and driving network edges and nodes on a map to understand the concept better visually.

As we discussed previously, we applied this network-based pathfinding on a hexagonal grid approach to visualize the spatial representation of accessibility. Accessibility analysis is commonly done through the use of distance-based spatial weight matrices. These matrices encode spatial relations between objects and can be used to define closeness, distance, position, and interaction. When using hexagonal global hierarchical geospatial indexing systems [21], such as the ones presented by h3 [22], we consider contiguity as sharing an edge, and that contiguous hexagons are equidistant in their central point [23]. We represent our map with hexagons, and

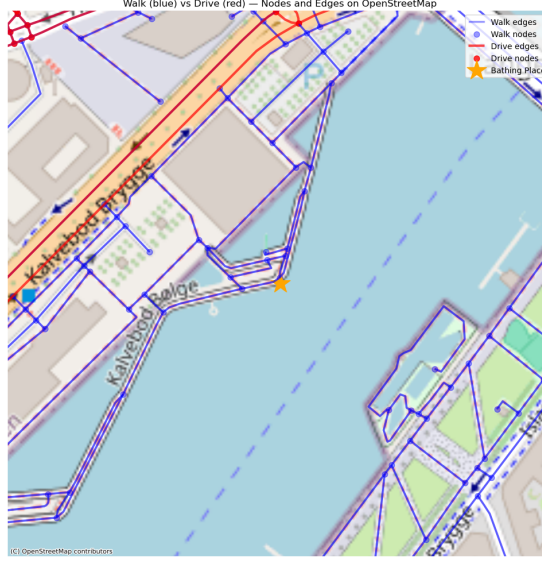


Figure 2: Visualization of networks

distance from points of interest as distance from these to the center of the hexagon. This generally works well, but we will show some cases in which the specific position has a deleterious effect on the distance to other places, due to geographical boundaries.

In order to build the backend of the application, we use *Flask*, a lightweight Python web framework. It allows us to create an API that supplies a map, the location of swimming places, and other information to the frontend of the system, which is based on HTML and JavaScript. The *networkx* library is used to handle the calculations on the network graph, enabling efficient shortest-path computations. *Pandas* is used to load and process the CSV dataset containing bathing place information, allowing us to filter and structure the data for further use.

Backend first loads the network graph in memory to access it later. After that, it extracts the geographic and contextual information for each bathing place from the dataset *final_dataset.csv*. Specifically, we retrieve the latitude and longitude coordinates for each location, which are essential for spatial analysis. In addition to these, we include relevant metadata as seen in Table 3 in Appendix A.

After data loading is complete, it renders the *map.html* with the data. Every time a user clicks on the map, the backend calculates distances to each swimming place and sends the data with the path for the closest one. It uses *shortest_path* function from *networkx* [24]. This function uses Dijkstra’s algorithm to find the shortest path in a graph. But to find the shortest path, the source and targets should be the nodes of the network. To achieve this we used function from *OSMnx* to get nearest node for both source and target but as it shown in the Figure 2 swimming places or the location we click on the interactive map might not have a node on it depending on the density and distribution of network nodes. That’s why it shifts to the nearest node, which affects the calculation. The consequence of this on the analysis will be discussed later in the discussion part.

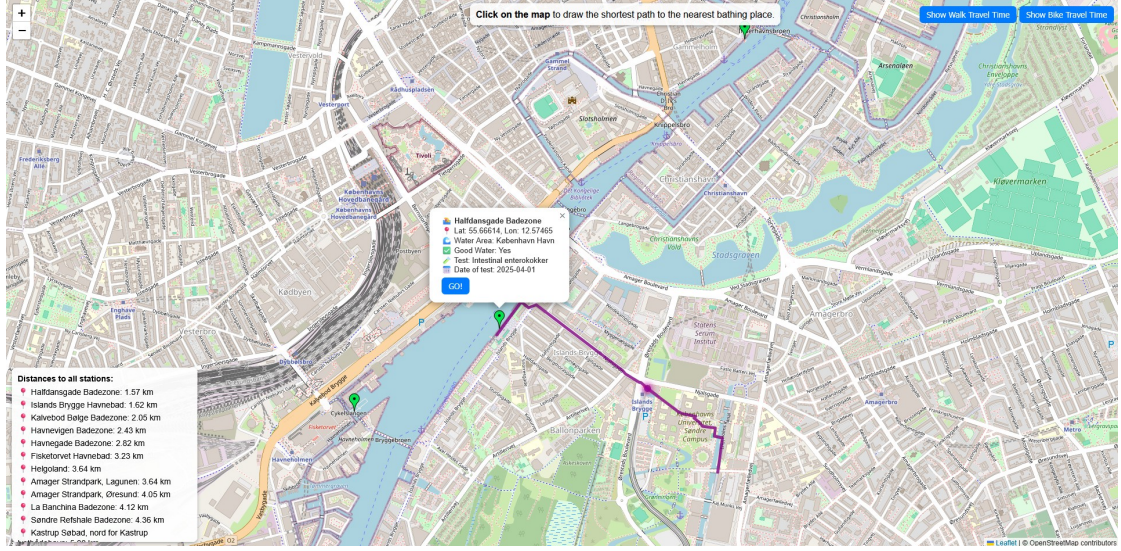


Figure 3: Shortest path from chosen point to closest bathing station

4.1 Hexagonal Isochrone Map Generation

The second part of our project involves generating hexagonal isochrone maps that visualize the accessibility of swimming places. For this, we developed *hexamap.py* script to create hexagons within the polygon that covers Copenhagen and that is predefined in a *geojson* file, and calculates the length of the shortest path to the closest swimming place for each hexagon using the same methodology that is described above. The only difference is that we defined the center of hexagons as the source location and then found the nearest node to the center on the network to calculate the shortest path. Since we have two hexagonal maps for both walking and biking, we took the same steps on both networks and generated two *geojson* files containing the polygons of hexagons and distance to the closest swimming place information. Results and possible misleading about that will be discussed later in the report.

5 Results

The result of our project is a prototype of a map of Copenhagen with different capabilities. First, the interactive map allows the user to pick any point and calculates the distance to all bathing stations in the city. It shows the path to the closest station, reachable by walking. As the map provides the distance to all the other swimming places, the user can choose any of them and press "GO!", which will generate the shortest path from the location to that swimming place.

Figure 3 shows the prototype. Having chosen a location in the city, ITU in this case, the map shows the shortest path to the closest bathing station, and a list of all the other stations in order of distance (bottom left). The pin represents the bathing station (green passing the water quality test, red not passing it), and by clicking on it, we get information about the current water quality measurements, geographical coordinates, and name.

On top of that, the map has an optional layer which shows the accessibility of the city in terms of walking 4a and biking 4b distance to the bathing stations. The hexagonal isochrone map categorises each hexagon in terms of how far (in minutes) it is from the closest bathing station.

The map shows the places further away from the waterfront to be red, and places closest to the water as blue, but it also highlights the areas around the water where there are no stations. On top of that, it shows the places that are geographically (or Euclidean) close to bathing stations, but that require a longer path to reach, due to urban obstacles, such as highways.

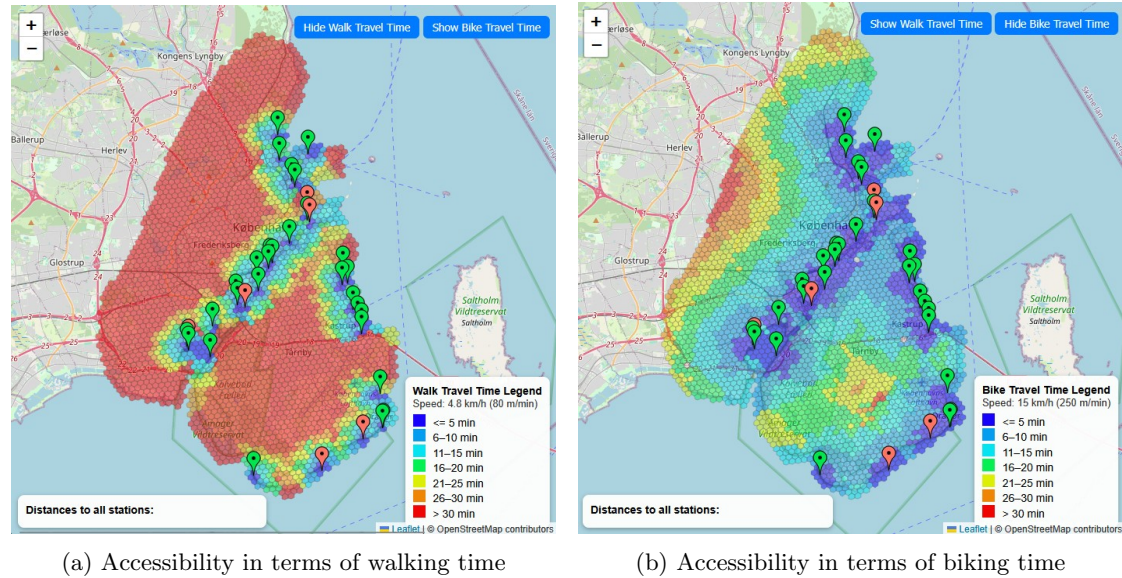


Figure 4: Accessibility maps of Copenhagen

In figure 4 we can see the change between the two maps, which is to be expected due to the difference in speed. In the top right corner, the two buttons allow the user to choose which map to show. Additionally, the user can still use the shortest path functionality on the map, allowing them to analyse the path from any colorful area.

It is relevant to note how some parts in the centre of the island are red, even when their close neighbours are green, which highlights the disparity in distance based on how the map provides paths. Additionally, some hexagons are without a colour, which is due to the lack of a biking path passing through the space.

As an interesting result of our accessibility map, we can see that the areas in blue and the population density (fig. 6 in Appendix C) of the city are quite similar in position. As presented in the background, the presence of amenities should be correlated with the amount of population, so that there is an even distribution of points of interest per amount of people. We can see from the map that swimming stations are visually distributed evenly in the city centre. However, we would need to do some more mathematical/geo-correlated analysis to be able to prove this empirically, which is currently out of the scope of the project.

There are two main challenges/drawbacks of our prototype. The first being, as mentioned in the data section, that some water quality measurements are quite old, compared to others. We hypothesize that it is due to the vicinity of other measuring stations, but we cannot be sure. On top of that, as the data was downloaded in April 2025, the latest water quality measurement is not up to date with the current time. Overall, however, the data source is considered to be trustworthy, so we prefer to have older measurements from this source than newer ones from other sources. Additionally, due to the way we are generating the map and the calculations we are doing, the prototype is quite heavy in terms of computational power. This is not optimal,

and deters us from being able to deploy the prototype, but it is acceptable as a proof of concept.

6 Discussion

The prototype is composed of two parts: a shortest path map from a user’s choice of location to the closest swimming place, and an accessibility map of the city. These two components are used to analyze the city of Copenhagen in terms of the current situation, to help users determine where to go swimming, based on water quality, vicinity, and ease of reach, and to determine how well the city is organized based on accessibility to bathing places.

The shortest path map we are showing only provides the walking routes, and is calculated based on nodes near the swimming stations. This way of calculating the route is much faster than other ways, but might be incorrect in terms of precise geographical location as we talked in Section 4. Another consideration for the shortest path generation is urban/natural obstacles, such as highways or swamps or lack of information such as lack of walking paths in network which exist in real life. For example, as shown in the Figure 5a near the ”Byskoven Badezone” swimming place, we can see that the hexagonal map has a red area very close. Analysing the shortest path provided, we can see that it takes a long detour, which we attribute to the lack of a path in the network we created by using OpenStreetMap. When we look at the same location in the bike network, we can see that there exists a path and it works as it should have been for the walking network (Figure 5b). This could be fixed by adding additional map sources, but it was out of scope for this project.

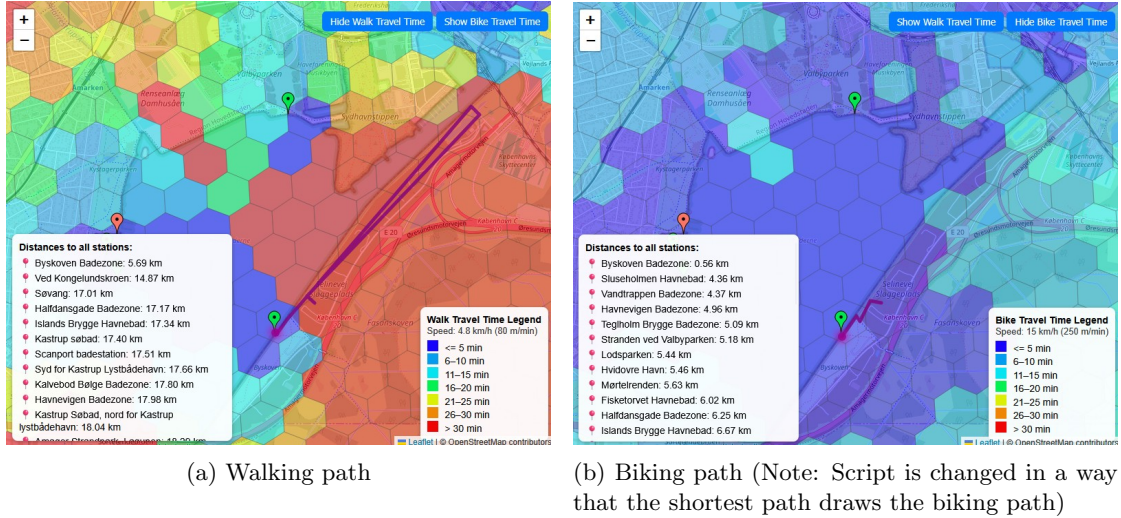


Figure 5: Problem of lack of route in the network

As mentioned in the Methods section (Section 4), when generating the accessibility map, we use the closest node in the network to the center of each hexagon to calculate the distance to swimming places. This approach is fairly accurate for walking maps but may be less precise for other modes of transportation. For example, when we use the biking network, since some swimming places are only reachable by walking, the map will show the closest place that a bike can reach. Similarly, if we were to take a public transportation stop, it might not be within the hexagon we are looking at, but in one near it, which would make our results the same for both hexagons, even if one is further from the station than the other

As mentioned in the results section 5, the prototype requires high computational power, which does not make it easy to use for everyday needs. However, once the map is calculated, and the accessibility map is shown, users can utilise it for assessing how reachable the swimming places are, for example, when choosing where they might want to move to. Additionally, when the city of Copenhagen considers new bathing stations to add, they might use the accessibility map to analyse the blind spots, which can support more informed decision-making. The bike map 4b shows that the majority of the city has reasonable access time to swimming stations, so they probably wouldn't need to invest great amounts of money to connect the entire city. On the other hand, we are calculating the map as the shortest distance to the closest station, which might not reflect the reality of a user choosing where to go swimming. Information such as the number of people, traffic, size of the bathing station, view, and preferred path might influence the decision, on top of water quality. We did not take any of them into consideration for this iteration of the project, but it could be a base for future work.

We have debated generating an additional accessibility map calculated with public transport travel time. Having it would allow for a more comprehensive evaluation of the reachability from the outskirts of the city to the centre, where the swimming places are mostly located. The travel time would decrease, and it would show a better analysis of the accessibility of the city with low-emission transportation systems. However, when generating the map, we noticed that OpenStreetMap does not allow us to choose the bus network when filtering for public transport, so we could only look at trains and metro networks, and stations, which cut off a major section of the public transportation system. This issue would have generated a map that is not faithful to reality, as the calculations for the closest swimming place from point X start from the closest node in the transportation network, which means the starting point would shift a lot. Future work could include a working solution to the problem by possibly adding different packages for bus networks or merging the transportation network with other networks.

7 Conclusion

The aim of the project was to create a prototype of a map of Copenhagen with two functionalities: a shortest path route from any chosen place in the map to the closest bathing station, and an accessibility map of Copenhagen based on walking and biking time.

We have provided an analysis of what we believe the use case for our product is, who can use it in which situation, and how it can be of use in the context of improving the city for better living conditions. While the map appears to match the population density map (Figure 6), there are other considerations to make when deciding how many and where to add new bathing places, which could include amount of people, size, water quality, and other amenities around the station, which would improve the experience of the user. We have discussed limitations of our product and future work in the discussion 6. We provide the prototype, which is accessible here: 

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Appendix A: Meta data tables

Software Metadata description	Details
Current software version	Python 3.12, libraries specified in requirements.txt
Github repo link	
Computing platform	The product should run on any system having more than 4GB RAM dedicated for this program. The 15km map is quite large, so expect some waiting time
Software License	MIT License
Software documentation and installation steps	README.md file in the repository [19]

Table 1: Software metadata

Field	Data Metadata Description
Data License	Free for use. Contains data used in accordance with the terms of use of Danish public data. https://miljoeportal.dk/dataansvar/vilkaar-for-brug/

Table 2: Data License

Column Name	Data Type	Additional Details
Name	String	Name of the bathing station
Latitude	Float (decimal degrees)	WGS84, EPSG:4326
Longitude	Float (decimal degrees)	WGS84, EPSG:4326
WaterArea	String	Body of water associated with the location,
Good water	Boolean	Indicates whether the water quality passed pre-defined criteria
Stofparameter	String	Type of chemical or biological test conducted
Dato	DateTime	Date of latest water quality measurement

Table 3: Description of final dataset used for maps

Appendix B: Contribution statement

The work was done as an equal collaboration of the three students. Each contributed with their knowledge and abilities, also considering the educational background of each. The data collection, analysis and cleaning was done by  focused on the code development of the prototype. The group agreed on the requirements, the capabilities and the specifics, and we individually spent the necessary time working on the different parts. Similarly the report. The content was agreed in plenum, so each student commented and agreed on what should be discussed, where to put the information and the style of the text and then each focused on different parts 

Appendix C: Additional Plots

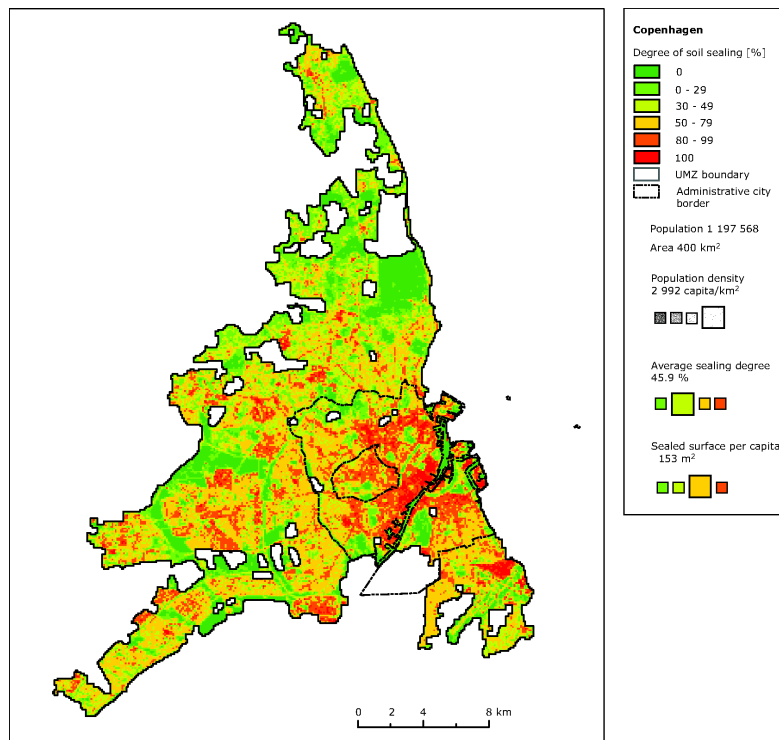


Figure 6: Population Density map [source](#)